

# CPSC 375

## Introduction to Data Science and Big Data Analytics

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**Week 01 L01**

# A bit about Me

Associate Professor

Ph.D., Computer Science,  
University of Oregon



# Introduction

- CSUF: Fall 2019
- Taught:
  - CPSC 120
  - CPSC 313
  - CPSC 315
  - CPSC 332
  - CPSC 375
  - CPSC 481
  - CPSC 483
  - CPSC 490
  - CPSC 499
  - CPSC 531
  - CPSC 589
  - CPSC 597
  - CPSC 598

# My Research Interests

- Model techniques for solver suggestions
- Study and analyze gender gap in STEM, CS
- Git repository analysis - HPC software
- Applying machine learning for various domains: cancer prognosis, etc
- Applying machine learning for solar cell reconfiguration

# A bit about yourself?

- In next 5 minutes:
  - Choose a partner
  - Introduce yourselves to each other
    - Learn your partner's name
    - Why they chose CPSC 375
    - Anything else they would like to share
- When done and asked
  - Tell your name
  - Introduce your new friend to the class

# What we will cover today

- Introductions
- What is Data Science?
- What is the relationship between Data Science and Big Data?
- Where does this course fit in CSUF?
- Course Outline
  - Canvas Tour
  - Syllabus

# Course Information & Prerequisites

- Course Number: CPSC 375: Introduction to Data Science and Big Data Analytics
- Units: 3
- Prerequisite:
  - a. CPSC 223P or CPSC 131
- Basic programming skills

# Office Hours



- Office: CS 530 (In-person and on Zoom)
- <https://fullerton.zoom.us/j/85718270183>
- Tuesdays and Thursdays: 2:30-3:30 PM
  - (In person and Zoom)
- Fridays: 12:00PM-1:00PM (In person only)

*If these times do not work for you, please send me an email and schedule an appointment time outside of the office hours. During final exam week, office hours are by appointment only.*



# How to reach me



- Kanika Sood
- Email: [kasood@fullerton.edu](mailto:kasood@fullerton.edu)
- Office Hours:
  - On Zoom (Link also on Canvas)
  - And In person

*If these times do not work for you, please send me an email and schedule an appointment time outside of the office hours. During final exam week, office hours are by appointment only.*

# Announcements

- Canvas
  - I don't check messages there, email me instead
  - **Unread for a long time**
- Questions?
  - Discussion Forums (Canvas and discord)
    - Any coding queries, deadline queries + ...
  - Email
    - Subject: **CPSC 375 + meaningful subject**
    - Turnaround time: ~ 2 working days **and** depends on your email subject
  - Office Hours

# Class Slides

A printable version of lecture slides will generally be available online (in Canvas) within 24 hours after each lecture

# Deadlines

- Canvas
- @ 11:59 pm on day of deadline -- unless time change specified

# Class Participations Points

- 5 % of overall grade
- Attendance **AND** In-Class Exercises
- Sign-up sheet

# Attendance

- Attendance is recommended
- Missing class is not advised unless it is excused by the instructor
- Miss a class
  - May lose points => class participation, discussion forum and in-class activities points
    - Class participation, Discussion forum in-class activities: 5 % of final grade
- Missing class as part of a documented accommodation is guaranteed to be excused
- The ADA accommodated student must make a reasonable effort to coordinate any absences with the instructor

# Class Logistics

- **CPSC 375-04**
- Room: EC-034
- Time: Fridays 2:00-4:45PM
  - Q&A: During the session
  - In class activities/exercises
- Please note: Up to 20 % of the classes can be online via Zoom. Zoom details will be shared before the online sessions.
- Canvas Class HomePage: Have access?

# Class Logistics

- Time:
  - Announcements
  - Lecture
  - Questions/surveys
  - In class exercises

## Break

- HW Instructions
- Lecture (contd.)
- In class exercises



# Class Logistics

- Time: 2hours 45 minutes
  - First 5 minutes: Announcements
  - Hr01:05- Hr02:15: Lecture/In class exercises
  - Hr02:15-Hr02:30 Break
  - Hr02:30-Hr03:40: Lecture/In class exercises
  - Hr03:40-Hr03:50: HW Instructions

# Textbooks & Syllabus

- [B1] [\*Python for Data Science\*](#), Arthur Turrell and [Contributors to Python4DS](#)
- [B2] [\*Learning Data Science\*](#), Sam Lau, Joey Gonzalez, and Deb Nolan
- [B3] [\*Mining of Massive Datasets\*](#), 3rd edition, Jure Leskovec, Anand Rajaraman, and Jeffrey D. Ullman
- [B4] [\*Introduction to pyspark\*](#), First Edition, Pedro Duarte Faria

Syllabus:

Available on Canvas - Check if you have access

# Homeworks

- 8-9 homeworks
- On Canvas
- 1-2 HW/week

# Exams

- MidTerm and Final Exam
- On Canvas
- Format: MCQ / Short Answers / True False / Fill in the blanks
- Midterm 25 %
- Final 25 %

# Assignment

- 25 % of final grade
- 8-9 Homework (HW) assignments
  - Written
  - Programming
- HW assignments: announced in class
- Due dates: announced in class

## Late HW:

- No credits
- Your responsibility: Hand in HW on time even when it is only partly completed.

# Class Project

- 20 %
- 2 submissions: Phase 1 and final
- Involves designing, implementing, and analyzing a substantial program in Python

# Grading

The final grade will be based on the following breakdown:

| Category             | % of Final Grade |
|----------------------|------------------|
| <hr/>                |                  |
| Homework             | 25 %             |
| Course participation | 5 %              |
| Projects             | 20 %             |
| Midterm              | 25 %             |
| Final                | 25 %             |

# Grading Scheme

In this course, the plus/minus system will be used. The grade breakdown is as follows:

- 98 – 100% = A+
- 93 - 97% = A (outstanding performance)
- 90 – 92% = A-
- 87 – 89% = B+
- 83 – 86% = B (good performance)
- 80 – 82% = B-
- 77 – 79% = C+
- 73 – 76% = C (acceptable performance)
- 70 – 72% = C-
- 67- 69% = D+
- 63 – 66% = D (poor performance)
- 60 – 62% = D-
- 0 – 59% = F

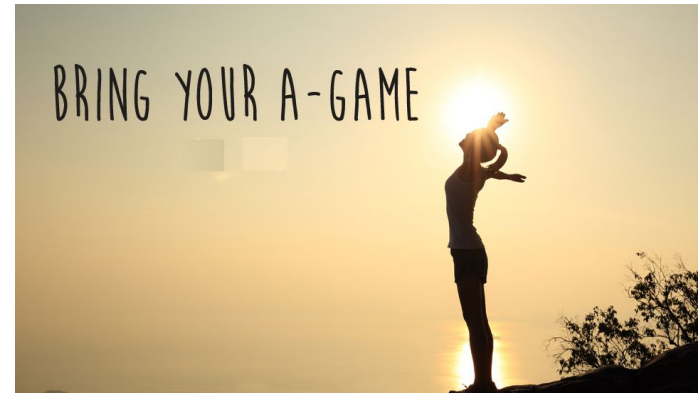
**Note:** No grades will be bumped higher than the assigned grades

The instructor reserves the right to adjust the grades up according to the performance of the whole class. However, there will not be any exceptions for any individual students.



# Class Expectations

- Lot of in-class exercises
- Lot of self learning/ self-coding
- Lot of hands-on guided through the textbooks



# Class Etiquettes

- Put your phones on silent
- Put your phones away - we will have breaks in class
- Stay on top of classwork



# Course Outline (Subject to change)

| Week   | Material                                                          |
|--------|-------------------------------------------------------------------|
| Week 1 | What is data science? Introduction to Python                      |
| Week 2 | Visualization                                                     |
| Week 3 | Exploratory data analysis                                         |
| Week 4 | Data wrangling                                                    |
| Week 5 | Tidy data                                                         |
| Week 6 | Linear modeling                                                   |
| Week 7 | Nearest neighbors classification<br>Distance measures             |
| Week 8 | K-means clustering<br>Evaluation metrics<br><b>Midterm Review</b> |
| Week 9 | Evaluation metrics (contd.)<br><b>Midterm exam</b>                |

# Course Outline (Subject to change)

| Week    | Material                                                 |
|---------|----------------------------------------------------------|
| Week 10 | Introduction to IoT; Time-series data (Intro.)           |
| Week 11 | Text data                                                |
| Week 12 | Text data (contd.)                                       |
| Week 13 | Data storage                                             |
| Week 14 | <b>Fall Recess (November 23 -29)</b>                     |
| Week 15 | Big data platforms (MapReduce)                           |
| Week 15 | Big data platforms (Apache Spark)                        |
| Finals  | <b><i>Final exam (covers material after midterm)</i></b> |

# No Use of AI tools permitted

This course assumes that work submitted for a grade by students – all process work, drafts, brainstorming artifacts, final works – will be generated by the students themselves, working individually or in groups as directed by class assignment instructions. This policy indicates the following constitute violations of academic honesty: a student has another person/entity do the work of any substantive portion of a graded assignment for them, which includes purchasing work from a company, hiring a person or company to complete an assignment or exam, and/or using generative AI tools (such as ChatGPT).

In this course, every element of class assignments must be fully prepared by the student. The use of generative AI tools for any part of your work will be treated as plagiarism. If you have questions, please contact me.

# Teaching Techniques

- 3 Credits: ? Hours of work?
- What happens when you walk through this door?
  - Mentimeter
  - Put your distractions away
  - Mental Breaks - in class

# Anonymous Survey

- ~ 5 minutes
- Use a code word and keep it handy till last week of classes
- First week:
  1. Week 1 survey: <https://forms.gle/72pABArzkn4y2gkY7>
- Last week

# Syllabus

- Available on Canvas
  - Check if you have access

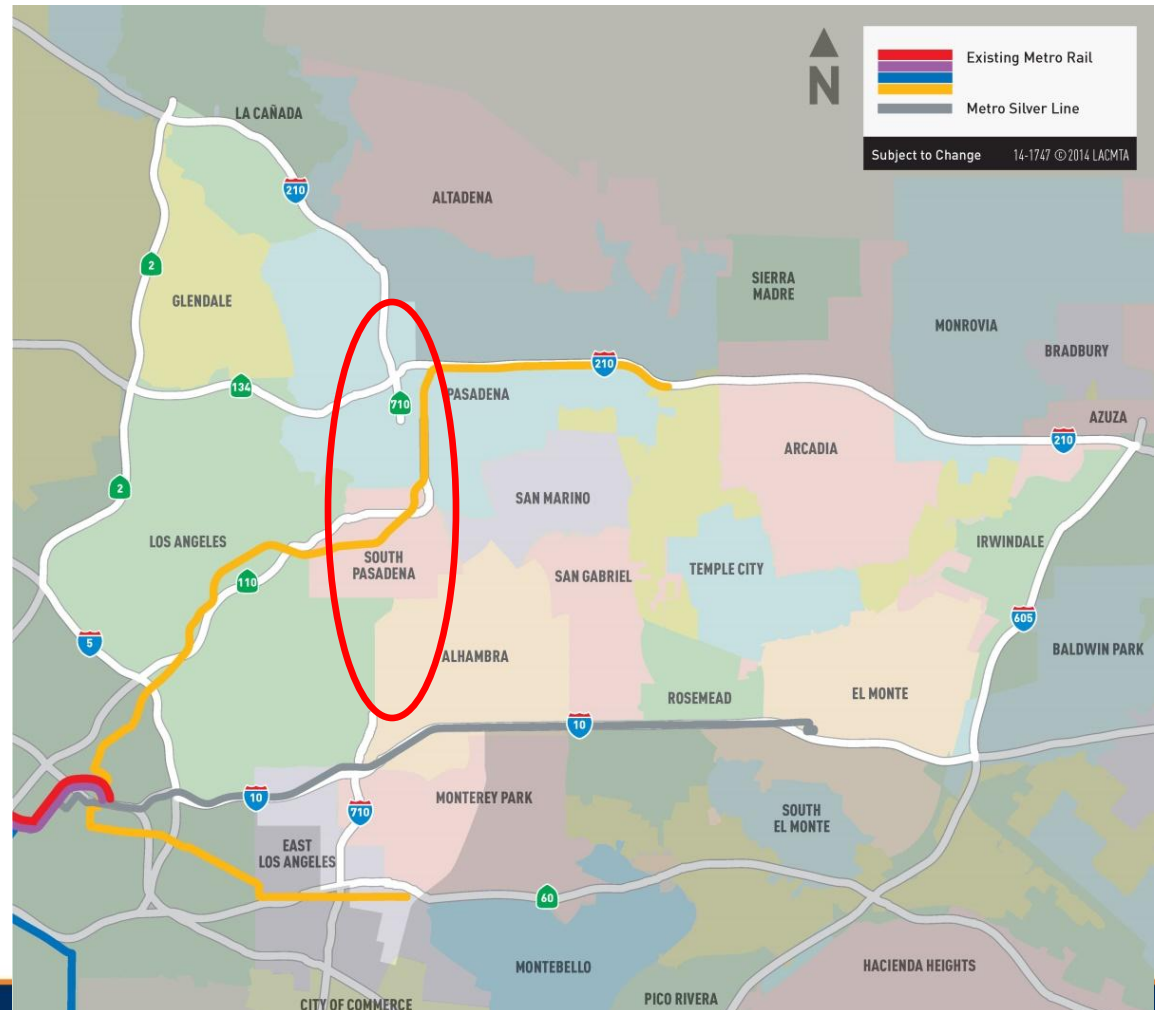


# Questions?

# Canvas Tour

# Example application

- I-710  
Freeway in  
Los  
Angeles



# Which of these are favored by the public?

- Some people **do not want** a freeway to go through their city
- Some people **want** a freeway to reduce traffic congestion in their city

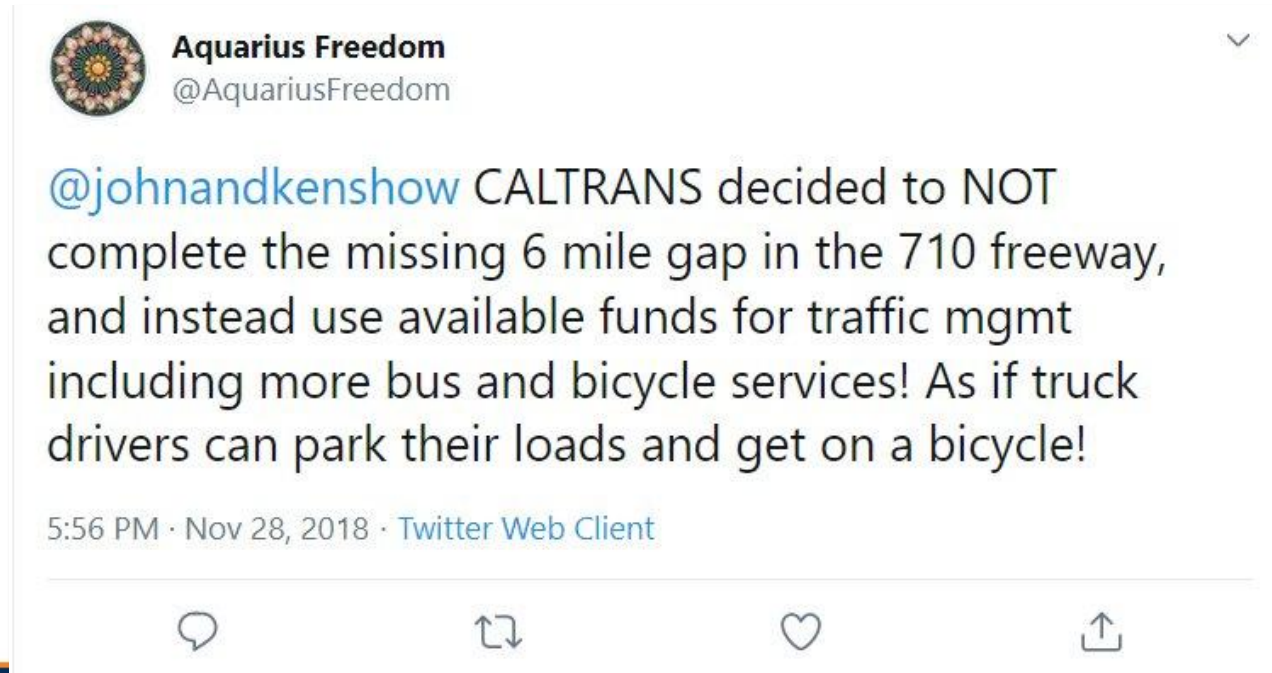
# 5 options before the public

- Formally developed by Caltrans

|   |                                                                              |
|---|------------------------------------------------------------------------------|
| 1 | • Bus Rapid Transit                                                          |
| 2 | • Freeway Tunnel                                                             |
| 3 | • Light Rail Transit                                                         |
| 4 | • No Build                                                                   |
| 5 | • Transportation System Management/Transportation Demand Management(TSM/TDM) |

# Use social media

- Analyze “sentiment” in Twitter tweets
- Sentiment: degree of subjective positivity/negative in a message



# Challenges

- Need to write programs to
  - Identify tweets that are about the 710 freeway project
  - Classify tweets into one of the 5 choices
  - Compute average sentiment for each topic/choice

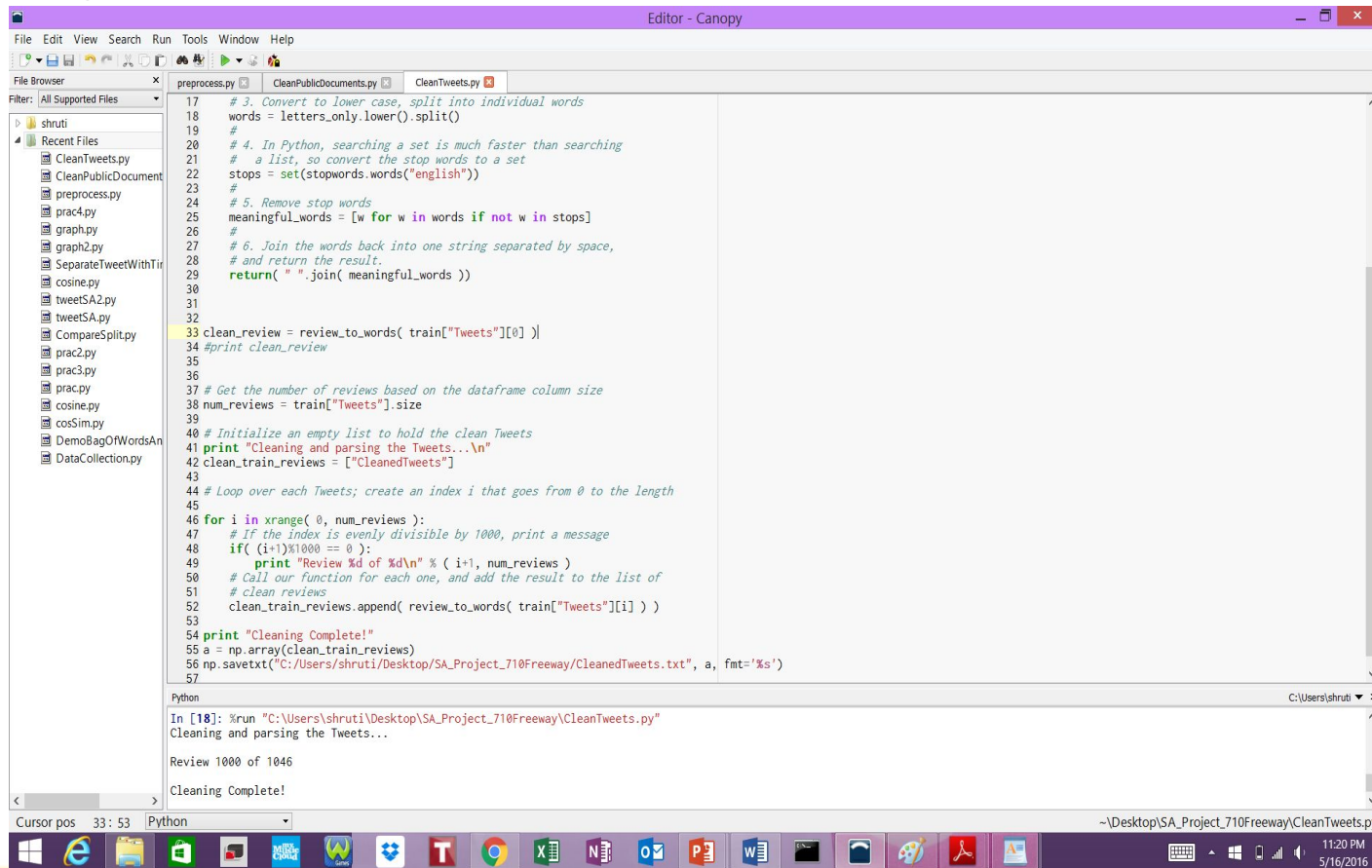
# Download data

```
Phrase
terrible man
seattle/big berth, boston's big dig, a new jersey to manhattan
tunnel, a 710 pasadena extension, all could be achieved by el
chapo
3:18 PM - 14 Jul 2015
San Bernardino News
Pasadena City Council opposes 710 Freeway tunnel: PASADENA >> The
City Council voted to oppose the 710 Freeway
12:00 AM - 15 Apr 2015
Foothill Gold Line
la canada flintridge pro-tem mayor donald voss says money for 710
tunnel extension should go to gold line foothill extension
instead.
10:56 AM - 25 Mar 2010
Hilda Solis
@KPCCSsharon McNary interviews Supervisor Solis re: community
concerns over 710N project.
11:41 AM - 21 Jun 2015
Alta Vista
Report supports building $5.65B #tunnel in #Calif. to cut traffic
congestion:
10:02 AM - 9 Mar 2015
Pasadena Star News
Alhambra hosts '710 Day' to advocate for 710 Freeway tunnel
10:37 PM - 10 Jul 2014
Catherine C.Stringer
@matthewstringer Rt @PasStarNews Alhambra hosts 710 Day to
advocate for 710 Freeway tunnel
1:16 PM - 11 Jul 2014
Pasadena Sun
Pasadena residents weigh 710 freeway extension options
6:15 PM - 2 Jun 2012
City of Commerce
I-710 Pavement Rehabilitation Project Full Freeway Closure:
9:29 AM - 16 Jul 2013
LA Daily Newz
Should a 710 #freeway #tunnel allow trucks & will it B safe?
9:31 PM - 31 Mar 2015
```



# Write (lots of) code!

- Pre-process the tweets



```
17 # 3. Convert to lower case, split into individual words
18 words = letters_only.lower().split()
19 #
20 # 4. In Python, searching a set is much faster than searching
21 # a list, so convert the stop words to a set
22 stops = set(stopwords.words("english"))
23 #
24 # 5. Remove stop words
25 meaningful_words = [w for w in words if not w in stops]
26 #
27 # 6. Join the words back into one string separated by space,
28 # and return the result.
29 return " ".join( meaningful_words ))
30
31
32
33 clean_review = review_to_words( train["Tweets"][0] )
34 #print clean_review
35
36
37 # Get the number of reviews based on the dataframe column size
38 num_reviews = train["Tweets"].size
39
40 # Initialize an empty list to hold the clean Tweets
41 print "Cleaning and parsing the Tweets...\n"
42 clean_train_reviews = ["CleanedTweets"]
43
44 # Loop over each Tweets; create an index i that goes from 0 to the length
45
46 for i in xrange( 0, num_reviews ):
47     # If the index is evenly divisible by 1000, print a message
48     if( (i+1)%1000 == 0 ):
49         print "Review %d of %d\n" % ( i+1, num_reviews )
50     # Call our function for each one, and add the result to the list of
51     # clean reviews
52     clean_train_reviews.append( review_to_words( train["Tweets"][i] ) )
53
54 print "Cleaning Complete!"
55 a = np.array(clean_train_reviews)
56 np.savetxt("C:/Users/shruti/Desktop/SA_Project_710\FreeWay\CleanedTweets.txt", a, fmt='%s')
57
```

Python

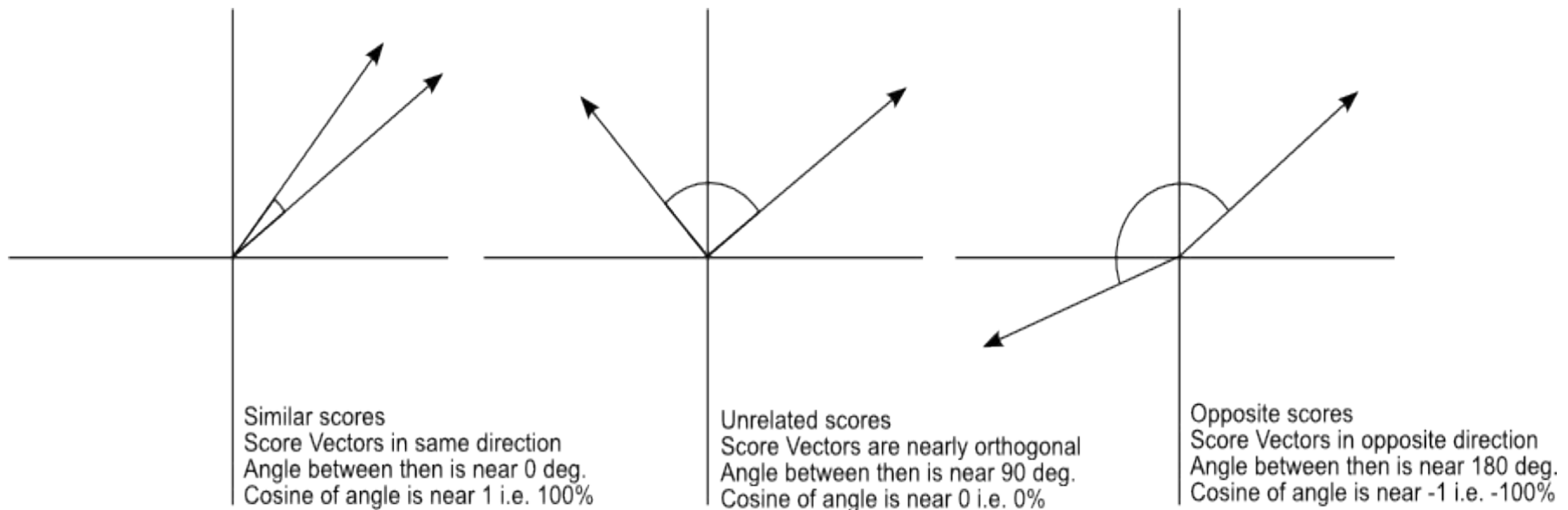
In [18]: %run "C:\Users\shruti\Desktop\SA\_Project\_710\FreeWay\CleanTweets.py"  
Cleaning and parsing the Tweets...

Review 1000 of 1046

Cleaning Complete!

# Compute similarity to topics

- Cosine similarity



- Write code for this

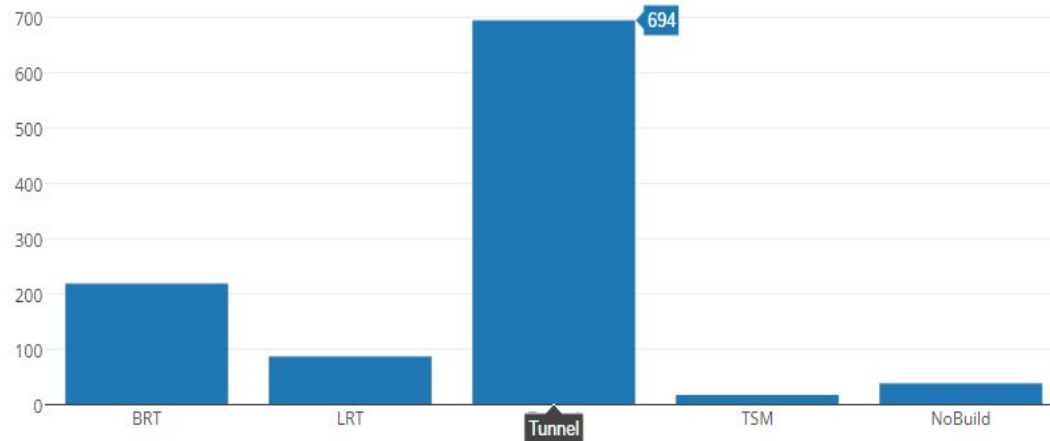
# Present results

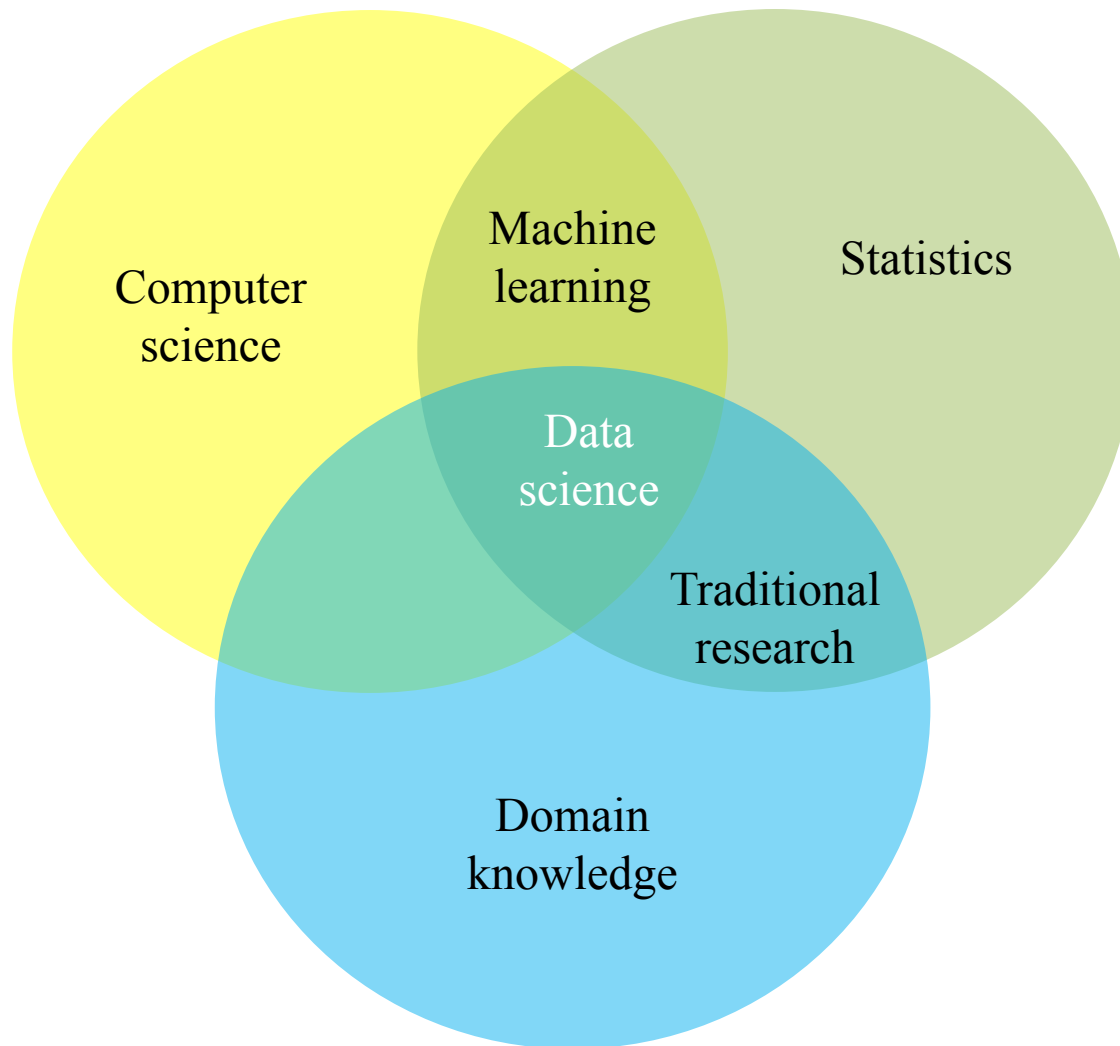
- Quantitative results

| Options  | Positive Tweets | Negative Tweets | Neutral Tweets | Total Tweets<br>(1046) | % Positives/<br>Total tweets |
|----------|-----------------|-----------------|----------------|------------------------|------------------------------|
| BRT      | 43              | 40              | 134            | 217                    | 4.11                         |
| LRT      | 14              | 7               | 64             | 85                     | 1.33                         |
| Tunnel   | 254             | 136             | 303            | 693                    | 24.28                        |
| TSM      | 5               | 3               | 4              | 15                     | 0.47                         |
| No Build | 6               | 6               | 24             | 36                     | 0.57                         |

# Present results

- Quantitative results





# Data Science – A Definition

**Data Science** is the science which uses computer science, statistics and machine learning, visualization and human-computer interactions to collect, clean, integrate, analyze, visualize, interact with **data** to create **data products**.

*data product*: an insight or tool created out of raw data that can be used to improve decision making<sup>+</sup>

<sup>+</sup><https://www.quora.com/What-are-data-products/answer/MoData>

# Examples of *data* -> *data products*

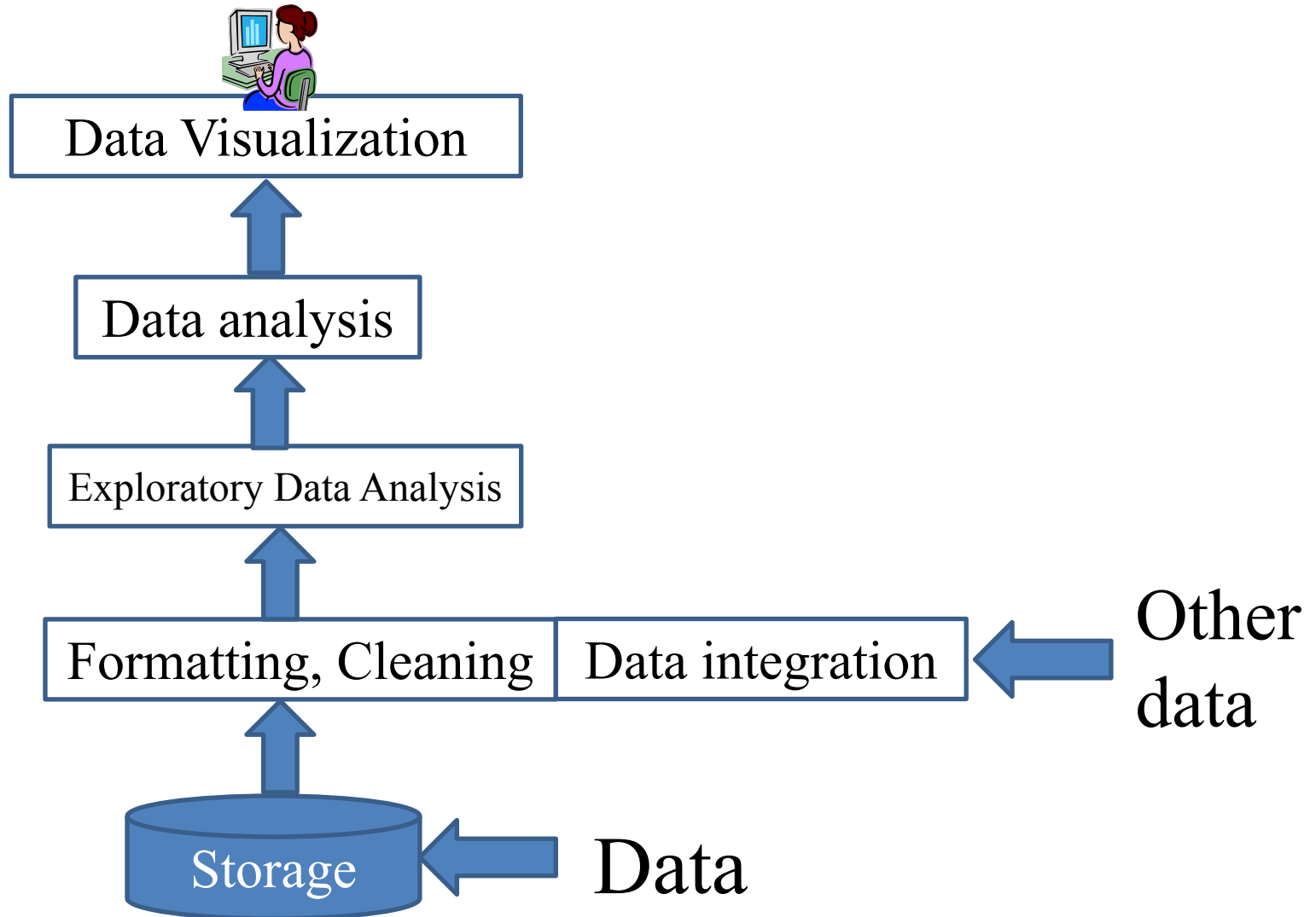
- Movie reviews -> movie recommendation (Netflix)
- Medical data -> diagnosis
- Smartwatch sensor data -> activity level (Fitbit)
- Tweets on Twitter -> [identify revolutions](#)
- Google search data -> flu activity [[Google Flu Trends](#)]
- Home sales data -> home selling strategy

# Class exercise

- In next 5 minutes:
  - With your classmate
  - Identify one data science application that you know
    - What is its “data”?
    - What is its “data product”?
- Present your application



# The Data Science Process



# Data science and Big Data

- What is “Big Data”?
- Just how *big* is big? Or is it just a relative term?
- What is the relationship between Big Data and data science?

# Big Data Everywhere!

- Lots of data is being collected and warehoused
  - Web data, e-commerce
  - Purchases at stores
  - Bank/Credit Card transactions
  - Social Network



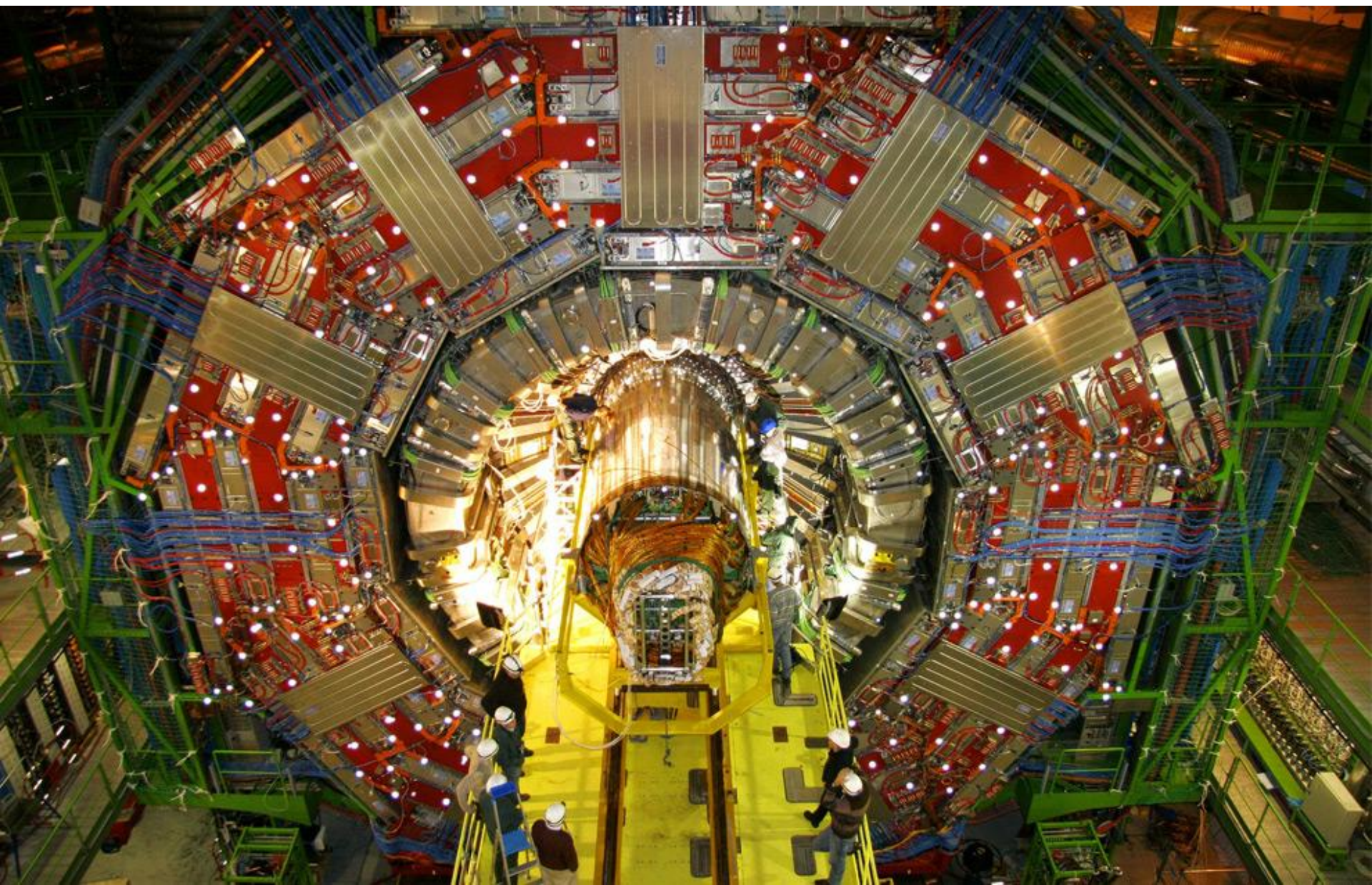
# How much data?

- CERN's Large Hydron Collider (LHC) generates 15 PB a year
- Oil and gas
  - "... a single well can produce terabytes of data in a couple of hours ..."



640K ought to be enough for anybody.





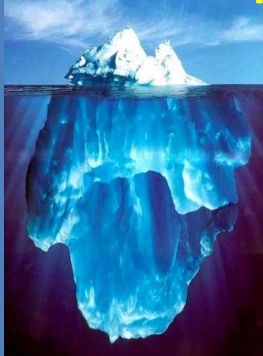
# 3 Vs of Big Data

- Raw Data: Volume
- Rate at which data comes in: Velocity
  - <http://www.internetlivestats.com/twitter-statistics/>
- Data types: Variety
  - Web clicks
  - Sensor data
  - Text: articles, tweets
  - Multimedia: Instagram, ...



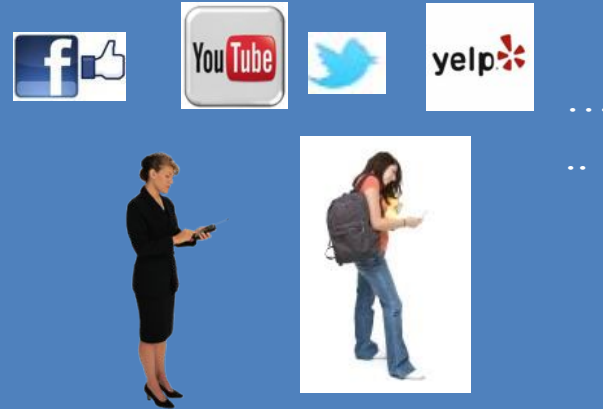
# Data Sources

## It's All Happening On-line



Every:  
Click  
Ad impression  
Billing event  
Fast Forward, pause,...  
Server request  
Transaction  
Network message  
Fault  
...

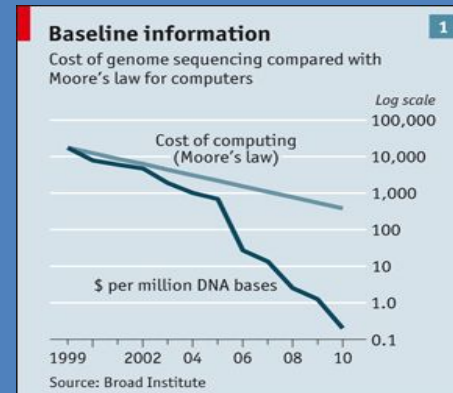
## User Generated (Web & Mobile)



## Internet of Things / M2M



## Health/Scientific Computing



# Context of the course at CSUF

- Counts as an upper division **elective** for CS majors
- Synergistic developments
  - [Big Data Discovery & Diversity](#)
  - [Courses on IoT in CS and Computer Engineering](#)



# Contents of the course

- Approximately 75% Data Science
  - Fundamental statistical analysis
  - Python programming language
- Other 25% Big Data
  - Focus on fundamentals
  - Most popular platforms
- Syllabus

# Thank you